



IFB452 Blockchain Technology Design Proposal

Student Details			
Student name	Rory Blair	Student number	N10773398

Instructions
You must use this template to complete your Design Proposal. Complete all the fields and then submit on Canvas using the instructions provided in both the assignment specification, and under “Assessment” on Canvas. The fields are not a fixed size for each question, so you can decrease their size, or expand them depending on your needs. Please do not modify the margins of this document. The font must remain at size 10 Arial, as per the directions in the page footer. Please do not remove any aspect of the template including the questions. Your report must be at between two (2) and (3) pages in length. Please note the page limit <u>excludes</u> this cover page, and any references. You may include graphics or screenshots in your proposal if it helps to convey your ideas. Before submitting this document, please save into PDF format. A brief oral presentation is required following submission of your proposal, followed by questions and answers from the teaching team. Marks for this assignment will be equally weighted between the written proposal and your comprehension demonstrated during the presentation. See the Rubric in Canvas for more details of the assessment criteria.

1. Proposed Application

1.1. Provide an overview of the blockchain application you propose to implement for your project, including the stakeholders involved and the types of transactions they will use.

Overview:

This blockchain will allow the tracking of military assets (gun, vehicles, ammunition) from their initial creation by the Manufacturer, through to the deployment, maintenance, transfer, disposal, and final fate of the inventory. This will allow for a tamperproof supply chain and ensure that diversion and loss of inventory are stopped.

Application:

The application shall be called 'MilitID' (Military ID). This blockchain application will be specifically used for identifying military assets and their supply chain history. A Token ID will be used for confirmation of the asset's history, with each Token ID being linked to the asset that exist in the real world. This linkage would be done via the asset's engraved serial number, which also allows the token to travel with the asset and its details through every stage of the supply chain.

This application doesn't necessarily need to be used for military inventory, as it could be used in other industries, such as the aviation or car industry, for tracking a vehicle and its details to ensure security.

Stakeholders:

- **Manufacturer** – The initial stakeholder who is accountable for the creation of the token and its initial details
- **Logistics Command** – Receive & assign assets to a military unit
- **Military Unit** – Deploy & use the asset whilst updating the status of the asset
- **Maintenance Facilities** – Records maintenance & Repairs
- **Decommissioning Units** – Execute final decommissioning & the assets' final status (fate).

Transaction Types:

- **Internal** – These transactions are controlled within the 'commission' & 'lifecycle' smart contracts.
- **Final** – Handled exclusively by the 'decommission' contract, in which the decommissioning units are the only stakeholders that can interact with this contract.

1.2 Provide a justification of why blockchain is a suitable technology solution for this application

Do You Need a Blockchain:

- **Do you need a Store State** – Yes → Immutable, tamper-proof record
- **Are there multiple writers** – Yes → 5 different stakeholders writing to the blockchain
- **Can you use an always online TTP** – No → should not have to be dependent on a single knowledge base that may fail at crucial times.
- **Are all writers known** – Yes → Should be national military or private military companies.
- **Are all writers trusted** – No → Although most stakeholders should be known, there are definitely real-world instances where national agencies and militaries have been impersonated or someone within has had their IDs compromised and used for malicious purposes.

Real-World Justification:

- **Importance:** Military inventory is a high-value asset and is subject to strict accountability.
 - **Loss/Modification of inventory:** Diversion of assets poses serious security threats, whilst untracked assets or 3rd party counterfeits could compromise assets.

- **Work already done:** Research conducted by Khalifa University validates that this application would be very beneficial for aerospace and defence supply chains [1]. Other work carried out by the U.S Department of Defence has tested blockchain systems as their traditional databases cannot guarantee an asset's detail when crossing multiple commands. [2].
- This approach enables the detection of inconsistencies, thereby preventing the diversion of assets into black markets [3].

2. Smart Contracts

2.1. Define the smart contract (s) you will use for your application, and discuss their functionality

Smart Contracts used in this application:

- **Commission** – This contract is responsible for creating the supply of tokens to be used within this supply chain and uses one function:
 - **Origin** – will create the entire supply of tokens
- **Lifecycle** – This contract manages the active lifecycle of the asset and utilises the following functions:
 - **Assign** – The Manufacturer connects the physical asset and its origin details to the token ID.
 - **Deploy** – Logistics Command assigns the asset to a Military Unit.
 - **Maintenance** – Maintenance Facilities update maintenance records.
 - **Transfer** – Allows assets to be reassigned from one Military unit to another.
 - **reportStatus** – Allows Military Units and Maintenance Facilities to lodge maintenance requests/outcomes, operational statuses, and location data.
- **Decommission** – This contract manages the final stage of the asset's lifecycle, which uses the following function:
 - **Fate** – Records the final status of the asset (Museum ship, destroyed, used, or scrapped) and is only accessed by the Decommissioning Units. This would be the final update made to that particular token as it is now out of active service.

Token Specific:

The token would be used to identify the type of asset as well as that asset's details and history. When looking up the token ID in the application, it would display the following details of that particular asset:

- ID
- Serial Number
- Manufacture Details (Date & location)
- Custody history
- Maintenance History
- Current Status (Location & condition)

2.2. Discuss interaction among the smart contracts in your application (if any) and the permissions of different stakeholders to read/write information on the blockchain

- The supply chain starts with the 'Commission' contract, where all the tokens that will be used are created, which can only be performed by authorised Manufacturers.
- The 'Lifecycle' contract is where the asset's active life is handled:
 - **Assign** – Only accessible by the Manufacturer's
 - **Deploy** – Only accessible by Logistics Command

- **Maintenance** – Only accessible by Maintenance Facilities
- **Transfer** – Only Accessible by Logistics Command
- **reportStatus** – Accessible by Military Units and Maintenance Facilities
- When the asset has reached the end of its lifecycle, the 'Decommission' contract is invoked:
 - **Fate** – Only accessible by Decommissioning Units and the final stage of the asset's lifecycle, thus preventing further modifications to that asset.
- All stakeholders are able to read the asset's full operational history, whilst writing access to the blockchain is strictly role-based.

3. Design considerations

3.1. Discuss the design considerations for your proposal

- **Start to End Traceability** – Every moment of an asset's lifecycle is recorded permanently, allowing stakeholders to assess if the asset is adequate for active duty. These records may also be used to see if problems had occurred with particular assets before.
- **High-Trust Stakeholders** – At critical stages of the asset's lifecycle, certain procedures can only be executed by reliable stakeholders. This is to ensure asset integrity is maintained by making sure the Manufacturer gives the correct asset details so that other stakeholders can be assured of the asset's reliability.
- **Non-Fungible Tokens (NFT)** – An NFT is another way to uniquely identify an asset and is usually used to prove ownership of said asset, whether digital or physical. With it being in the blockchain, the NFT is tamper-resistant and a perfect way for identifying military assets [4].
- **Possible integration with other military systems** – Radio Frequency Identification (RFID) tags, which are utilised in the field today, usually report back to a legacy system, which would be out-of-date. This blockchain could also integrate the purpose of that legacy system into the blockchain, with the RFID tags automatically updating the status of the asset [5].

3.2. Discuss how you will address these considerations in your implementation. Include any limitations for your design

- Address these considerations:
 - **Start-to-End Traceability** – This is already built into the design, as it is a crucial part of the blockchain process.
 - **High-trust Stakeholders** – This will be coded in as certain stakeholders being granted permission-based roles to write specific details to the blockchain.
 - **NFTs** – This type of token is excellent as it perfectly connects a token ID with a physical asset.
 - **Military system integration** – It would be a slow but possible option, which should be explored after the blockchain is created and verified to be working properly.
- Limitations
 - **Classified Information** – Not all assets would be on the blockchain, as some governments would still have secret agendas and not record certain assets that would be used.
 - **NFT Generation** – At the moment, token generation will be a problem as only a finite number of tokens can be created and used.
 - **Integration** – will be difficult to integrate legacy systems and procedures into this application. Would also need to train stakeholders in how the application works.
 - **Laws & Regulations** – Some countries have strict rules and regulations on the use of NFT's and the transportation of military assets.

4. References (excluded from page limit)

4.1. List any references in this section.

- [1] R. W. Ahmad, H. Hasan, I. Yaqoob, K. Salah, R. Jayaraman and M. Omar, "Blockchain for aerospace and defense: Opportunities and open research challenges," *Computers & Industrial Engineering*, vol. 151, 2021.
- [2] M. T. Simerly and D. J. Keenaghan, "Blockchain for military logistics," U.S Army, 4 November 2019. [Online]. Available: https://www.army.mil/article/227943/blockchain_for_military_logistics. [Accessed 21 March 2026].
- [3] J.-P. Beaudet, "Immutable Security: How Blockchain Strengthens Military Supply Chains," The Digital Chamber, 1 December 2025. [Online]. Available: <https://digitalchamber.org/how-blockchain-strengthens-military-supply-chains/>. [Accessed 21 March 2026].
- [4] Australian Government, "Non-fungible tokens," Australian Government, 23 June 2025. [Online]. Available: <https://www.ato.gov.au/individuals-and-families/investments-and-assets/crypto-asset-investments/transactions-acquiring-and-disposing-of-crypto-assets/non-fungible-tokens>. [Accessed 19 March 2026].
- [5] NSTXL, "Blockchain for Military Logistics: What You Should Know," NSTXL, 4 April 2023. [Online]. Available: <https://nstxl.org/blockchain-for-military-logistics/>. [Accessed 21 March 2026].